



Does biochar affect soil wettability and flow pattern?

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ARTICLE INFO

Handling Editor: Budiman Minasny

Keywords:

Water repellency
Biochar
Biochar–soil mixture
Non-zero contact angle
Unstable flow

ABSTRACT

Despite the known benefits of biochar application to soils, little is known about its effect on soil wettability and the consequences on soil water movement across the soil profile. Focusing on the latter, three types of commercial biochar derived from wheat, corn, and rice straw produced under low temperature (350–450 °C) were mixed with a wettable sandy soil at rates of 0%, 2%, 5%, and 10% w/w. Soil water repellency (SWR) of the biochars and biochar–soil mixtures was determined by water drop penetration time (WDPT) test and sessile-drop contact angle (CA) measurement. The WDPT results (< 5 s) showed complete wettability for the biochar–soil mixtures, whereas CA results indicated some water repellency. The mean initial CA of the three biochars varied between 105.3° and 113.9°, with ~ 30° for the pure sand, and between 56.8° and 75.7° for the biochar–soil mixtures. SWR increased with biochar-application rate, whereas biochar particle size (< 1 mm and > 1 mm) had no significant effects on SWR. A 2-D flow chamber experiment with point source water application at the surface was used to continuously monitor the flow pattern and soil water content (SWC) distribution across the soil profile during infiltration and redistribution periods. Biochar–soil mixtures (2% and 5% w/w) with the three biochar types were studied. Flow chamber results showed substantial differences in plumes shape and internal SWC distribution in the biochar–soil mixtures compared to the untreated sand during the wetting and drainage. The plumes' shape and nonmonotonic spatial SWC distribution suggested unstable flow in the biochar–soil mixtures. The internal fingers developed in the biochar treated sand indicate that SWR induced by biochar addition seemed to be physically induced via the blending of the biochar and soil particles, rather than chemically induced, via coating of the soil particles with amphiphilic molecules. Under field conditions, the primary and inner finger-like plumes may eventually form preferential flow paths that will affect the spatial distribution of water and fertilizer in soil profiles and their availability to plant roots.

1. Introduction

Biochar, first reported in dark soil (*terra preta*) from the Amazon Basin (Sombroek, 1966), refers to the solid organic material resulting from conversion of biomass by thermochemical decomposition under oxygen-free or limited-oxygen conditions (Meyer et al., 2011). In recent decades, the industrial production of biochar and its land application for soil amelioration and long-term carbon sequestration have raised widespread concern in the scientific community (Lehmann, 2007; Woolf et al., 2010; Sohi, 2012). Given its unique physicochemical properties and recalcitrant nature, land application of biochar necessarily leads to a series of irreversible and long-lasting ecological and environmental consequences. Therefore, a comprehensive understanding and careful

evaluation of these potential consequences must be achieved before its extensive land application.

To date, many studies have addressed the short- and long-term effects of biochar on soil chemical and biological processes. Most have reported its beneficial influence on soil fertility and crop productivity (Jeffery et al., 2011; Biederman and Harpole, 2013), nutrient leaching and heavy metal availability (Laird et al., 2010; Rees et al., 2014), soil microorganism, fauna and plant root activity (Lehmann et al., 2011), soil organic carbon sequestration, and mitigation of non-CO₂ greenhouse gas emission (Lehmann, 2007; Woolf et al., 2010; Cayuela et al., 2014). Comparatively, less attention has been paid to its effects on the soil's physical properties and water flow in biochar-treated soils (Atkinson et al., 2010; Blanco-Canqui, 2017). Studies addressing these

Abbreviations: SWR, Soil Water Repellency; WDPT, Water Drop Penetration Time; CA, Contact Angle; SWC, Soil Water Content.

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<https://doi.org/10.1016/j.geoderma.2022.115826>

Received 11 October 2021; Received in revised form 7 March 2022; Accepted 10 March 2022

Available online 22 March 2022

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effects have focused mainly on soils' static physical properties, such as structural properties (e.g., bulk density, soil porosity, and aggregate stability), soil–water retention indices (e.g., field holding capacity, plant-available water content, and permanent wilting point), and soil hydraulic properties (e.g., saturated and unsaturated hydraulic conductivity, and steadiness of the infiltration rate) (Abel et al., 2013; Herath et al., 2013; Hardie et al., 2014; Peak et al., 2014; Castellini et al., 2015; Ojeda et al., 2015; Novak et al., 2016; Obia et al., 2016). Although different types of biochar were applied at various rates to different soil types in those studies, the consensus was that biochar application can affect soil structure, soil hydraulic properties, and consequently, soil hydrological processes. Furthermore, the high porosity and specific surface area of biochar were identified as being responsible for biochar's diverse effects on soil porosity (Hardie et al., 2014). However, another important physical property of biochar, the hydrophobicity of its organic surface, may also influence soil hydrological processes by causing soil water repellency (SWR) (Kinney et al., 2012; Ganz et al., 2013; Jeffery et al., 2015; Chen et al., 2021).

Soil water repellency (SWR) is the phenomenon whereby a soil does not wet spontaneously when in contact with water (DeBano, 2000; Doerr et al., 2000). This has significant effects on hydrogeomorphological processes by reducing infiltration capacity (Wallis et al., 1990; Imeson et al., 1992), increasing surface runoff and erosion (Doerr et al., 2003; Ferreira et al., 2016), decreasing capillary rise and local water availability, and forming an unstable, irregular wetting front and finger flow (Bauters et al., 2000; Xiong et al., 2012; Ganz et al., 2013; Rahav et al., 2017), which in turn accelerates the leaching of nutrients and contaminants to the groundwater (Blackwell, 2000; Ogunmokin and Wallach, 2021). It is commonly accepted that SWR is closely associated with soil organic matter, which in nature is subject to the dynamic balance between input from plant litter and root exudate and output from microbial decomposition. The two main groups of organic compounds responsible for SWR—the nonpolar aliphatic hydrocarbons and polar substances with amphiphilic structure (Doerr et al., 2000; Horne and McIntosh, 2000)—are mainly from plant residues (McGhie and Posner, 1981; Mao et al., 2016), soil microorganisms (Rillig, 2005; Feeney et al., 2006), and the substances produced in carbon-cycling processes (Valat et al., 1991). However, a considerable amount of organic matter can also be introduced into the soil by human activity, such as organic fertilizer application, effluent irrigation, and organic soil amendments. This artificially imported organic matter can cause SWR directly due to its hydrophobic properties, its coating of soil particles, or simply its presence in the pore spaces of the soil matrix (McGhie and Posner, 1981; Verheijen et al., 2010; Aranda et al., 2016; Liu et al., 2019). Indirectly, it can affect SWR by promoting microorganism activities that produce hydrophobic compounds, such as n-alkanes, olefins, and monoketones (Doerr et al., 2000).

In a critical review, Verheijen et al. (2010) pointed out a gap in knowing the mechanisms of biochar induced SWR and the consequent effects on soil hydrological processes. Several studies on pure biochar have recently related its physicochemical properties to hydrophobicity. The results confirmed the hydrophobic nature of biochar and the dependence of that hydrophobicity on the amount of residual labile aliphatic functional groups remaining on the biochar surface after pyrolysis (Zornoza et al., 2016; Useviciute and Baltreinaite, 2020). Some authors have stated that biochar produced at high temperatures has the highest hydrophobicity due to aromaticity (Cantrell et al., 2012; Al-Wabel et al., 2013; Bubici et al., 2016), whereas other studies reported that biochar hydrophobicity decreases with increasing pyrolysis temperature and disappears when the peak pyrolysis temperature reaches a threshold level (Kinney et al., 2012; Das and Sarmah, 2015; Zornoza et al., 2016). The type of feedstock and pretreatment procedures may also play an important role in affecting the degree of biochar hydrophobicity (Kinney et al., 2012; Zornoza et al., 2016).

Rather than looking at the hydrophobicity of pure biochar, information on the water repellency of biochar–soil mixtures would be more

valuable for an evaluation of the environmental effects of biochar application. The persistence and intensity of SWR are usually quantified by water drop penetration time (WDPT) and soil–water interface contact angle (CA), respectively. A soil is generally considered to be water-repellent when it has a WDPT > 5 s or a CA > 90° (Doerr et al., 2000). Kinney et al. (2012) reported that the WDPT of a sandy soil increased from < 1 s to 61–870 s after 2%–7% biochar application. Other studies concluded, based on WDPT tests, that there is no significant decrease in wettability for a soil mixed with strongly hydrophobic biochar (Abel et al., 2013; Herath et al., 2013; Baronti et al., 2014; Saha et al., 2020). Yet other studies reported that CA values of soil–biochar mixtures were < 90° (Smetanová et al., 2013; Eibisch et al., 2015; Ojeda et al., 2015; Mao et al., 2019). However, the clear increase in CA after biochar application must be noted in these studies, because a slight increase in CA will have a remarkable effect on soil water movement, even if it is < 90° (Aminzadeh et al., 2011; Wallach et al., 2013; Wang and Wallach, 2020). In a two-dimensional (2-D) flow chamber experiment, unstable finger flow was observed at a CA of 75°, and the water flow changed from stable to unstable when the CA increased from 48° to 56° (Wallach et al., 2013). Thus, it is reasonable to argue that the application of hydrophobic biochar to wettable soil will affect soil water movement by increasing the degree of SWR. Using biochar produced from peanut hulls at a temperature of 500 °C, Githinji (2014) found a significant decreasing trend of cumulative infiltration with increasing rate of biochar application in a sandy loam soil and attributed it to the strong hydrophobicity of the biochar. Similarly, Jeffery et al. (2015) reported high hydrophobicity (CA ~ 118°) of biochar produced from herbaceous feedstock at 400–600 °C, and postulated that the application of this biochar would have negative effects on water retention and field saturated hydraulic conductivity of a wettable sandy soil. Edeh and Masek (2021) revealed a significant reduction in wettable soils' water-retention capacity and hydraulic conductivity following the application of hydrophobic biochar. However, the effects of biochar application on the precise soil water movement across the soil profile during infiltration and drainage processes have never been directly studied. Therefore, the current study aimed to reveal the change in water repellency of wettable soil caused by biochar application and the consequent effects on soil water movement within the soil profile.

2. Materials and methods

2.1. Soil and biochar materials

The soil used in the current study was taken from an experimental field site in a commercial citrus orchard in Sitiriyia (31°54'07"N, 34°51'23"E) located in the central part of the Israeli Coastal Plain. Disturbed mineral soil samples were taken from the 0–5 cm surface soil layer using a hand trowel between adjacent tree rows from the alley. This area was chosen as it does not receive any salinity and fertilizers by irrigation water, which might affect the soil properties. Before the laboratory experiments, the wettable rainfed soil was air-dried and passed through a 0.5-mm sieve. The soil was sandy, with 84% sand, 2% silt, and 14% clay. Organic matter content was 1.79%, and initial water content was around 0.84% after air-drying.

Three types of commercial biochar (BC) were used in the current study. One was made from wheat straw (BC-W) by LiuHe Qinfeng Straw Technology Company in China's Jiangsu Province. The other two biochars were made from corn (BC-C) and rice straw (BC-R), respectively, by Sanli New Energy Company, Henan Province, China. The pyrolysis temperature of these three biochars was between 350–450 °C, as is common in commercial biochar production.

The specific surface area of the biochar materials was about 218.2 m² g⁻¹, 197.6 m² g⁻¹, and 254.3 m² g⁻¹ for BC-W, BC-C, and BC-R, respectively, determined by BET-N₂ absorption method (Quadrascorb SI, Quantachrome Instruments, Boynton Beach, FL, USA). The microstructure of the sandy soil and biochar particles was imaged using a

scanning electron microscope (JSM IT-100, JEOL, Tokyo, Japan).

The purchased biochar mass was obtained in particulate form. For uniform mixing with the soil particles, the biochar was ground and passed through a 2-mm sieve followed by oven-drying at 105 °C for 24 h. The biochar was then mixed with the soil to homogeneity for SWR measurements without incubation at three application rates of 2%, 5%, and 10% by weight. The application rates were chosen to cover the literature's broad range of application rates (Jeffery et al., 2011; Blanco-Canqui, 2017). The BC-W was further sieved into two particle size groups: fine (< 1 mm, BC-Fine) and coarse (> 1 mm, BC-Coarse). Finally, the BC-Fine and BC-Coarse were mixed with soil particles at application rates of 2%, 5% and 10%, respectively.

The surface chemistry of the biochar was determined by Fourier-transform infrared (FTIR) spectroscopy (Nicolet 6700, Thermo Fisher Scientific, Waltham, MA, USA). The biochar samples were scanned from 400 cm^{-1} to 4000 cm^{-1} at 2-wavenumber resolution. The particle size distribution of the soil, biochars and soil–biochar mixtures was determined by laser diffraction particle sizing using a Mastersizer 2000 (Malvern Panalytical, Malvern, UK).

2.2. Measurement of SWR

Before carrying out the SWR test, the soil, biochars, and biochar–soil mixtures were again oven-dried for 72 h at 55 °C and then cooled under constant laboratory conditions of 24 °C and ~ 50% relative humidity for 3 h.

2.2.1. WDPT test

The persistence of SWR was quantified by the WDPT test, which consists of placing a drop of distilled water on the soil surface and measuring the time to its complete penetration (Doerr et al., 1998). The soil and biochar–soil mixtures were packed in a cylinder (100 cm^3) with a controlled bulk density of 1.243 g cm^{-3} , close to the soil bulk density measured in the field during soil sampling. On the other hand, the packed biochar only reached a bulk density of 0.800 g cm^{-3} . Three 50- μL drops of distilled water were placed on the exposed material (soil, pure biochar, and biochar–soil mixture) surface, and the time elapsed before each drop was completely imbibed was read from a video recorded by a high-speed camera. Soils are classified as water-repellent if the WDPT exceeds 5 s (DeBano, 2000; Doerr et al., 2000). SWR is generally divided into five classes according to WDPT values: class I soil is wettable, not water-repellent (WDPT < 5 s); class II is slightly water-repellent (5 s ≤ WDPT ≤ 60 s); class III is strongly water-repellent (60 s < WDPT ≤ 600 s); class IV is severely water-repellent (600 s < WDPT ≤ 3600 s); and class V is extremely water-repellent (WDPT > 3600 s) (Bisdorf et al., 1993).

2.2.2. CA measurement

The intensity of SWR is quantified by the CA between the soil–water interface and the water–air interface using the sessile drop method. The CA for water-repellent soils usually decreases with degree of repellency due to rearrangement of the organic molecules coating the soil particle surfaces (Doerr et al., 2000). The initial CA and its variation with time were measured with a goniometer (EasyDrop DSA20E, Krüss GmbH, Hamburg, Germany). The material (soil, biochars, and biochar–soil mixtures) was glued to a double-sided adhesive tape attached to a microscope slide to form a flat single-particle layer (Bachmann et al., 2000). A 10- μL water drop was used for all sessile drop measurements. A high-resolution digital camera recorded the dynamic drop shape every 1 s up to 100 s, which was then mathematically described by adopting the Young–Laplace equation for curved boundary areas.

Three slides were made for each tested material, and 5 droplets were placed at different locations on each slide. The average value of the 15 droplets was calculated to represent the temporal variation of CA. In general, a material surface with an initial CA value > 90° is considered hydrophobic.

Differences among mean CA values of different treatments were tested using least significant difference (LSD) method at 5% probability level. The statistical analyses were performed using SPSS software (version 20.0, IBM Corp., Armonk, NY, USA).

2.3. Flow chamber experiment

The flow chamber experiment allowed for continuous monitoring of the flow pattern and soil water content (SWC) distribution across the soil profile during the infiltration and drainage periods without any destructive sampling (Wallach and Jortzick, 2008; Xiong et al., 2012; Wallach et al., 2013; Bughici and Wallach, 2016; Ogunmokin et al., 2020). The experimental setup consisted of a transparent flow chamber, 20 cm wide, 22.5 cm high, and 0.5 cm thick (inner dimensions), packed with pure soil or biochar–soil mixtures (three types of biochar with two biochar-application rates—2% and 5% w/w) and a digital camera to record the infiltration and drainage processes at different water-application rates. The front and back walls (25 cm wide, 25 cm high, and 0.5 cm thick) of the chamber were connected through two side rods (22.5 cm wide, 2.5 cm high, and 0.5 cm thick) and one bottom rod (25 cm wide, 2.5 cm high, and 0.5 cm thick) with screws, leaving the top open. The chamber was packed uniformly with great care to avoid layering and to achieve a uniform soil bulk density throughout the chamber. As a result, a consistent bulk density of 1.243 g cm^{-3} was achieved for all examined biochar–soil mixtures. The outer surfaces of the back and front walls were cleaned carefully to obtain a clear view of the soil profile inside the chamber after packing. The chamber was then fixed to a clamp mounted to the wall to ensure adjustment of the cell to the same position for each run, and leveled vertically and horizontally with a spirit level. The chamber volume was carefully selected to allow for the wetted plume to adequately expand during wetting and redistribution without resulting in air entrapment within it.

Two diffused light-emitting diode (LED) boxes (50 cm wide and 100 cm high) were used to illuminate the flow chamber, located on the left and right sides of the camera–flow chamber axis (170 cm from the center of the flow chamber). Uniform diffused light intensity was achieved on the front face of the flow chamber by adjusting the angle of the lightbox faces toward the chamber. In addition, the LED boxes' light intensity was adjusted to ensure maximum distinction (in terms of gray value) between dry and wetted soil.

A high-speed, high-resolution monochrome camera (Zyla sCMOS-5.5-CL10, Andor, Belfast, UK) with a zoom lens (AF Nikkor, 24–85 cm, 1:2.8–4D, Nikon, Tokyo, Japan) was used to quantify light reflection from the flow chamber. The resolution of the camera sensor was 2560 × 2160 active pixels with dimensions of 6.5 × 6.5 μm . The camera was placed on a tripod (190 cm away from the chamber's center) behind the LED boxes to avoid the camera's reflection on the chamber. The center of the lens was adjusted to the same height as the center of the chamber. For this camera setup, 3585 pixels on the image represented 1 cm^2 of the soil profile. The frames from the camera were loaded and saved as 8-bit images on the computer using Andor's camera control and analysis software platform (SOLIS version 4.28.3).

Water was applied as a point source on the soil surface at three flow rates: 0.5 ml min^{-1} , 2 ml min^{-1} , and 5 ml min^{-1} through a 0.3-mm (inner diameter) Tygon pipe connected to a peristaltic pump (Masterflex L/S, Cole Parmer Instrument Company, Vernon Hills, IL, USA). The outlet pipe was located 0.5 cm above the middle of the soil surface to prevent the effect of drop kinetic energy and water burst on water penetration. The total volume of applied water for all runs was 10 ml. Following the wetting stage, water was allowed to redistribute for an additional 10 h. For each run, an image of the chamber filled with dry soil was taken prior to water application as a reference for the subsequent image-analysis process. During the wetting and redistribution phases, images were taken every 5 s to enable high-frequency monitoring of the water-flow dynamics. Each treatment run was repeated at least twice, with similar results for each repeat.

Digital analysis of the images was performed with the “Image Processing Toolbox” of MATLAB software (version 7.11, MathWorks, Natick, MA, US). The input images were a matrix of 2560×2160 Gy values for each image. For each flow chamber, an initial gray-value matrix of the dry soil-biochar mixture substrate was taken as a reference before water application. This matrix was subtracted from the gray-value matrices taken during the experiment to eliminate the effects of minor deviation from uniform illumination and spatial soil heterogeneity that might be erroneously referred to as changes in SWC. Given that the relationship between the gray value and volumetric moisture content is linear (Wallach and Jortzick, 2008), the slope of the linear calibration curve was determined by the water mass balance (Wallach et al., 2013). The slope was calculated as the ratio between the sum of the changes in water content at each pixel at a specific time and the sum of the changes in gray value at every pixel over the entire chamber. Since the initial value of the dry soil image was subtracted from subsequent images, it was assumed that the calibration line emanates from the axis origin. The slope for each run was determined for each image during wetting and redistribution periods. Representative results are shown in section 3.3.

3. Results

3.1. Biochar particle size and FTIR

The scanning electron microscopy images of the three types of biochar and the sandy soil are shown in Fig. 1. The biochar particles' porous characteristics (Fig. 1a–c) differed from that of the mineral sandy soil particles (Fig. 1d). Fig. 2 shows that the biochar had predominantly finer particles ($\sim 80\%$ between 2 and 100 μm), while the mineral soil particle size was larger ($\sim 75\% > 100 \mu\text{m}$). The particle size distribution did not change much for the 2% ($\sim 70\% > 100 \mu\text{m}$) and 5% ($\sim 65\% > 100 \mu\text{m}$) biochar mixed with sandy soil (Fig. 2). The particle size distribution for BC-C, BC-W and their mixture with soil particles were similar to the

results of BC-W in Fig. 2 (data not shown). Given the size of the biochar particles relative to the mineral soil particles (Figs. 1 and 2), it is clear that the finer biochar particles become trapped in the intrapore spaces of the soil particles upon mixing, rather than coating the mineral soil particles to form one entity. The spectrum characteristics of the three biochars are shown in Fig. 3. Peaks in the $2800\text{--}3000 \text{ cm}^{-1}$ C–H stretch region revealed the presence of aliphatic functional groups on the biochar surface.

3.2. Water repellency measurement

The WDPT values for the sandy soil, biochars, and biochar–soil mixtures were < 5 s (data not shown); therefore, all were classified as wettable. Notably, for the biochar itself, since the material is light and puffy, the small dry biochar particle climbs up onto the water drop surface, and the biochar matrix caves in immediately when the water drop is placed on it. However, according to CA measurements and the 90° criterion, the three biochars (mean initial CAs of 106.1° , 105.3° and 113.9° for BC-W, BC-R, and BC-C, respectively) were classified as water-repellent, whereas the 2%, 5%, and 10% (w/w) biochar–soil mixtures were subcritically water-repellent (Fig. 4). The CA for the biochars decreased for about 20 s and then became relatively stable after 40 s, with CA values $> 65^\circ$ (Fig. 4). The initial CA of the pure soil was much lower than those of the pure biochars and decreased rapidly from an initial CA of 35.4° to 0° within 10 s. The initial CA of the biochar–soil mixtures relative to the pure soil increased by 60.5%, 88.7% and 113.8%, reaching 56.8° , 66.8° and 75.7° for 2%, 5% and 10% mixtures, respectively (Fig. 4). The rate of CA decrease for the biochar–soil mixtures was much lower than for the pure soil—decreasing rapidly during the early stage (< 10 s) and stabilizing at $\sim 30^\circ$, $\sim 40^\circ$ and $\sim 60^\circ$ for the 2%, 5%, and 10% mixtures, respectively (Fig. 4). These results indicate that the commercial biochars are strongly water-repellent, and their mixtures with wettable soil are subcritical water-repellent.

The influence of the biochars' particle size on the CA is shown in

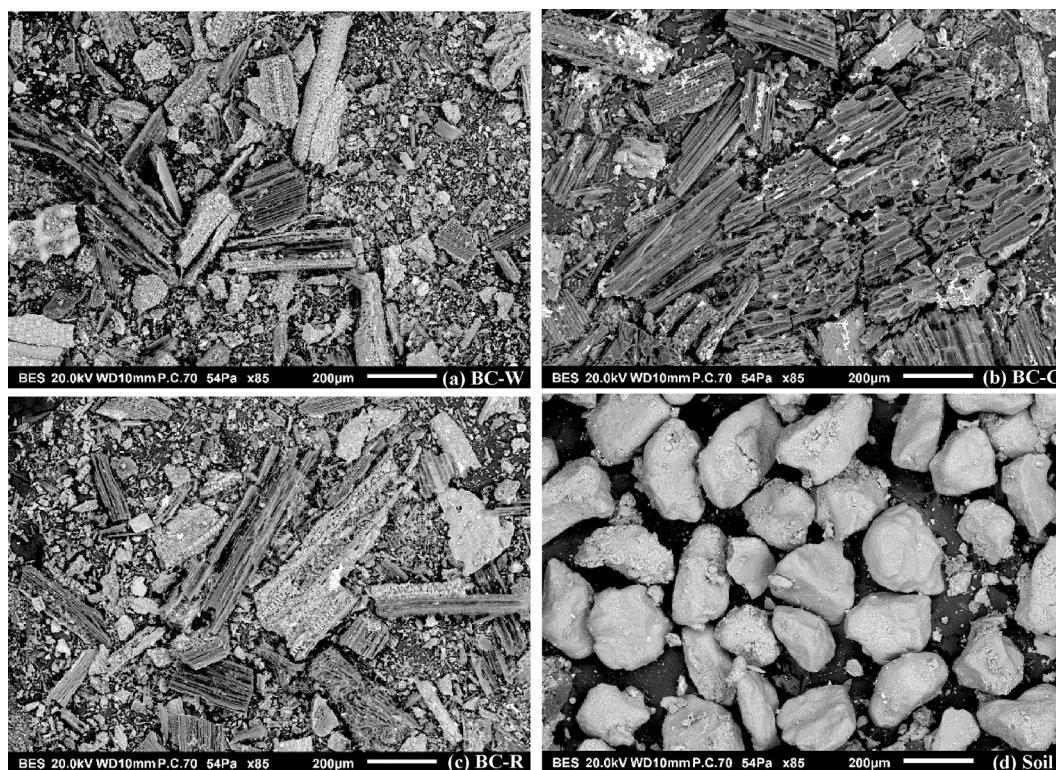


Fig. 1. Scanning electron microscopy (SEM) images of the (a) BC-W, wheat straw biochar; (b) BC-C, corn straw biochar; (c) BC-R, rice straw biochar; and (d) sandy soil at $\times 85$ magnifications.

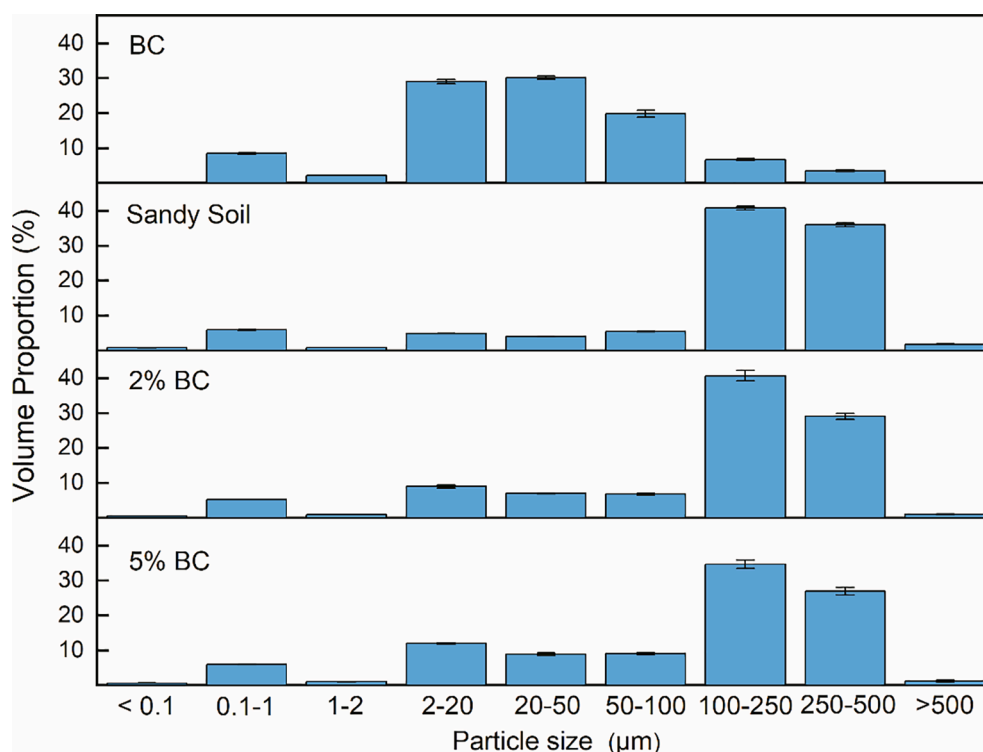


Fig. 2. Particle size distribution of the biochar (BC-W), sandy soil and soil mixed with 2% and 5% biochar.

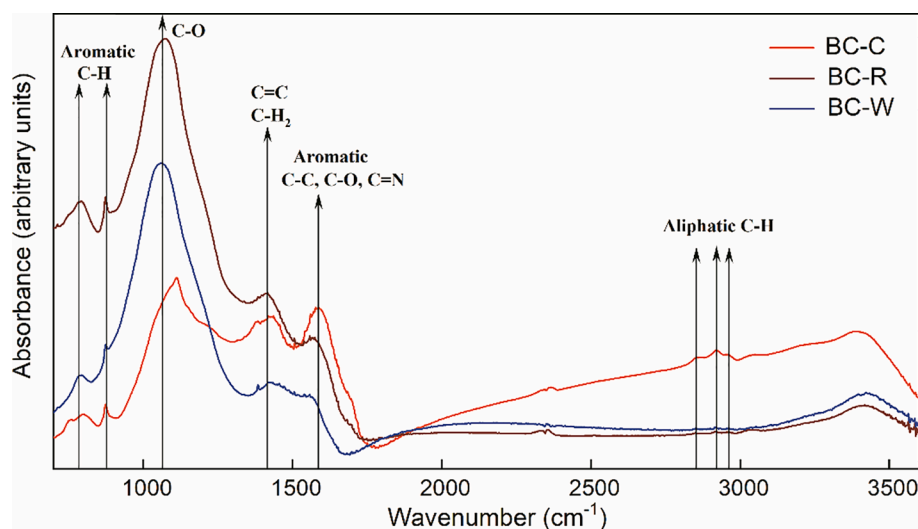


Fig. 3. Fourier transform infrared spectroscopy (FTIR) spectra of the commercial biochar derived from corn straw (BC-C), rice straw (BC-R) and wheat straw (BC-W) at the pyrolysis temperature around 350–450 °C.

Fig. 5. The initial CA was 112.2° and 100.2° for BC-fine and BC-coarse particles, respectively, indicating strong water repellency. The CA of BC-Fine (< 1 mm) was significantly higher than that of BC-Coarse (> 1 mm), and there was no significant difference in CA between BC-Fine and the biochar that included all particle sizes (BC). The CA of biochar–soil mixtures significantly increased with the biochar-application rate for BC, BC-Fine, and BC-Coarse applications. For a 2% mixing rate, there was no difference in the initial CA between BC and BC-Fine mixtures, whereas the CA for soil and the BC-Coarse mixture was relatively lower ($P < 0.05$). In contrast, there was no significant difference in initial CA among 5% and 10% mixing rates and all biochar particle sizes. These results indicate that biochar particle size has no significant effect on the SWR of biochar–soil mixtures.

3.3. Flow chamber study

The 2-D flow chamber enabled examining the biochars' effect on plume shape and the temporal variation in SWC dynamics within the plume during the wetting and subsequent redistribution stages (Figs. 6–8 show results for BC-W). The plume size for the 0.5 ml min^{-1} water-application rate decreased during the wetting stage, while the overall water content increased as the biochar mixing percentage increased (Fig. 6a). In addition, the plume width at the soil surface decreased with biochar percentage. In contrast to the gradual decrease in SWC with radial distance from the water source in the sandy soil, the SWC was uniform within the plume with an abrupt decrease through a thin layer along the plume perimeter in the 2% biochar–soil mixture

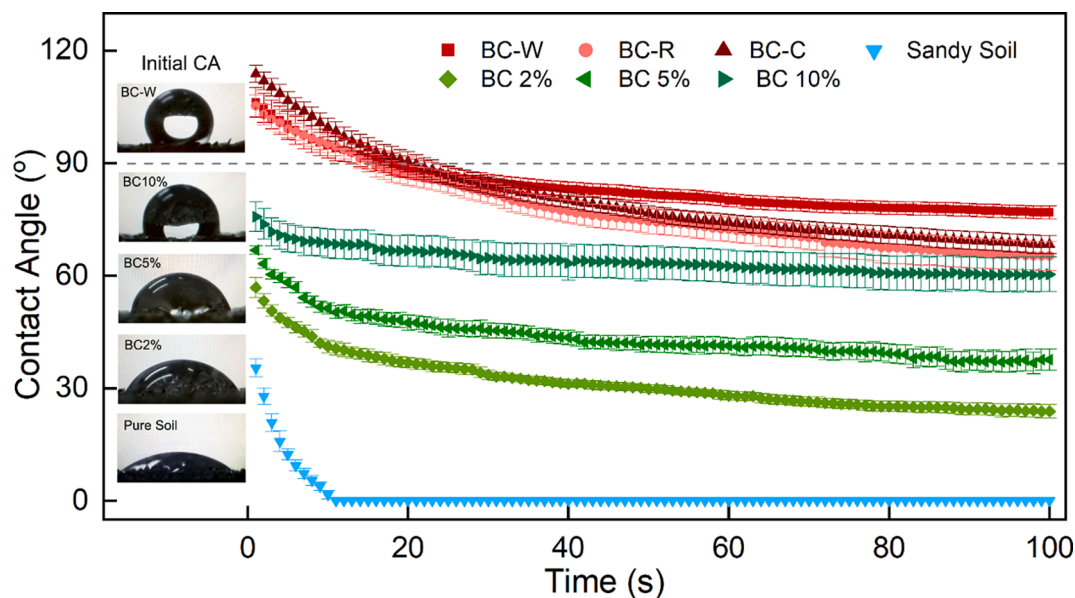


Fig. 4. Contact angle measurements of sandy soil, three types of biochar, and soil mixed with 2%, 5% and 10% (w/w) biochar. The CA of the biochar-soil mixtures are the average for the biochars.

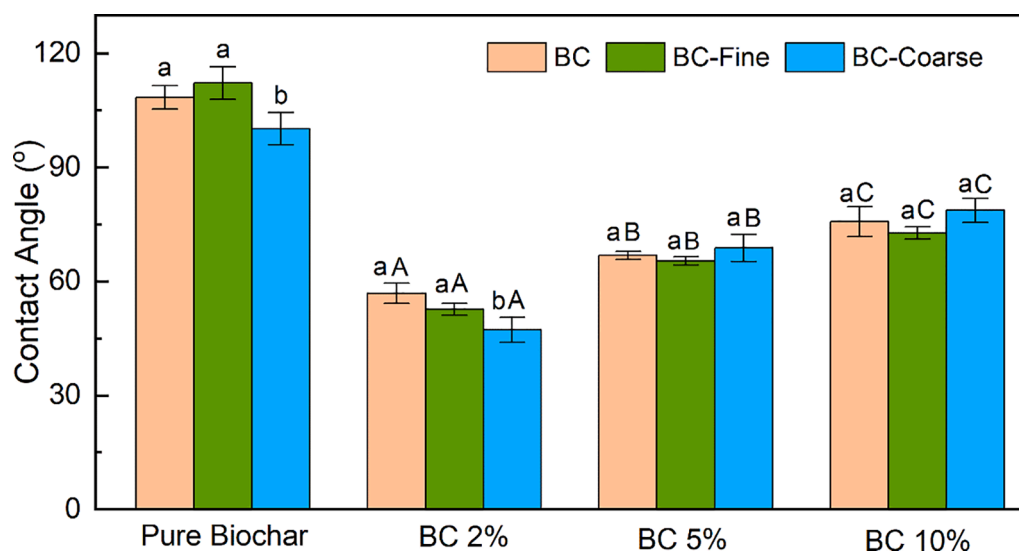


Fig. 5. Initial contact angle of biochar (BC-W) with different particle size and their mixture with soil at 2%, 5% and 10% (w/w) biochar application rate. The results are similar for BC-C and BC-R and their mixtures with the sandy soil.

plume (Fig. 6a). In the 5% biochar-soil mixture, the SWC was higher than for the other two biochar rates. As a result, internal vertical fingers developed within the plume (Fig. 6a).

The plume's expansion for the 0.5 ml min^{-1} water-application rate during redistribution relative to its shape at the end of the wetting stage was lower with increasing percentage of biochar (Fig. 6b). The SWC redistribution within the plume was uniform in the sandy soil as compared to the 2% and 5% biochar-soil mixtures, and the boundary between the end of wetting and start of redistribution was visible in these mixtures. Moreover, the fingers within the plume persisted through the redistribution phase.

The differences in plume size and internal SWC distribution observed for the 0.5 ml min^{-1} water-application rate (Fig. 6) were generally repeated for the two higher water-application rates (2 ml min^{-1} and 5 ml min^{-1}). The plume size in the sandy soil became smaller with higher internal water content as the biochar-application rate increased (Figs. 7 and 8). A core with higher SWC was obtained below the water-

application point surrounded by a wide dry soil contour. For the 2% and 5% biochar-soil mixtures, substantially wider plumes formed with a shallower depth. The internal SWC during the wetting stage increased with water-application rate and biochar percentage in the mixture. Under a 2 ml min^{-1} flow rate, the wetting stage's SWC distribution within the plume was more uniform for the 2% and 5% biochar treatments than for pure soil. For both water-application rates, the wet core was surrounded by a narrow strip of dry soil along the plume perimeter (Figs. 7 and 8). Remarkably, a rough finger-like flow pattern developed at the lower boundary of the inner high SWC area for the 5% biochar-soil mixtures under 2 ml min^{-1} and 5 ml min^{-1} water-application rates (Figs. 7 and 8). Notably, the internal fingers developing in the biochar-soil mixtures for the 2 ml min^{-1} application rate were less apparent than those developing at the low (0.5 ml min^{-1}) water-application rate (Fig. 6).

Results similar to those in Figs. 6–8 were obtained in the replicate experiments using BC-C and BW-R (Fig. 9).

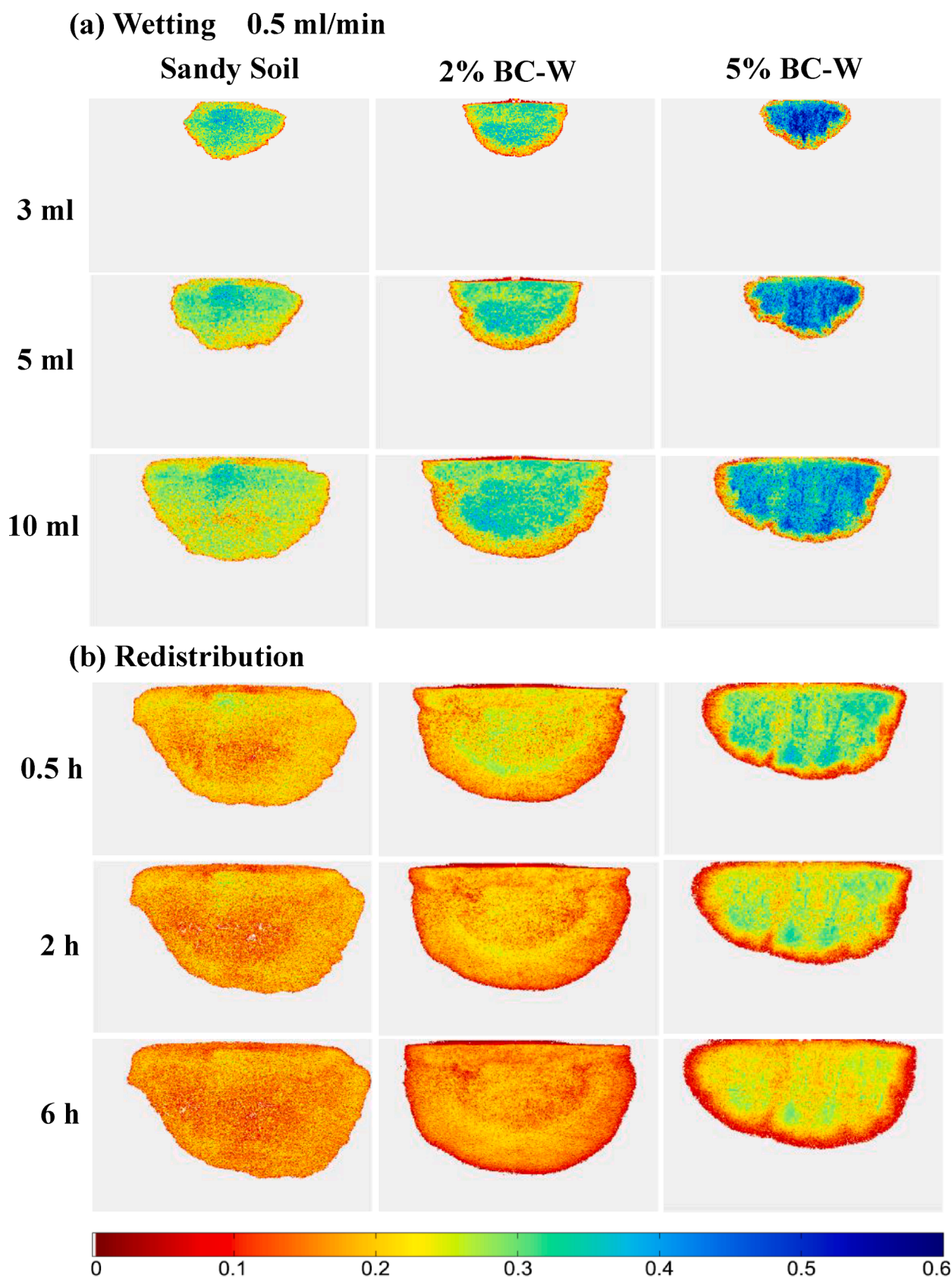


Fig. 6. Soil water content distribution during (a) wetting, and (b) redistribution in sandy soil and soil mixed with 2% and 5% wheat straw derived biochar (BC-W). Water was added to the soil surface as a point source at a flow rate of 0.5 ml/min. Moisture content varies from red (dry) to blue (wet) according to the color bar.

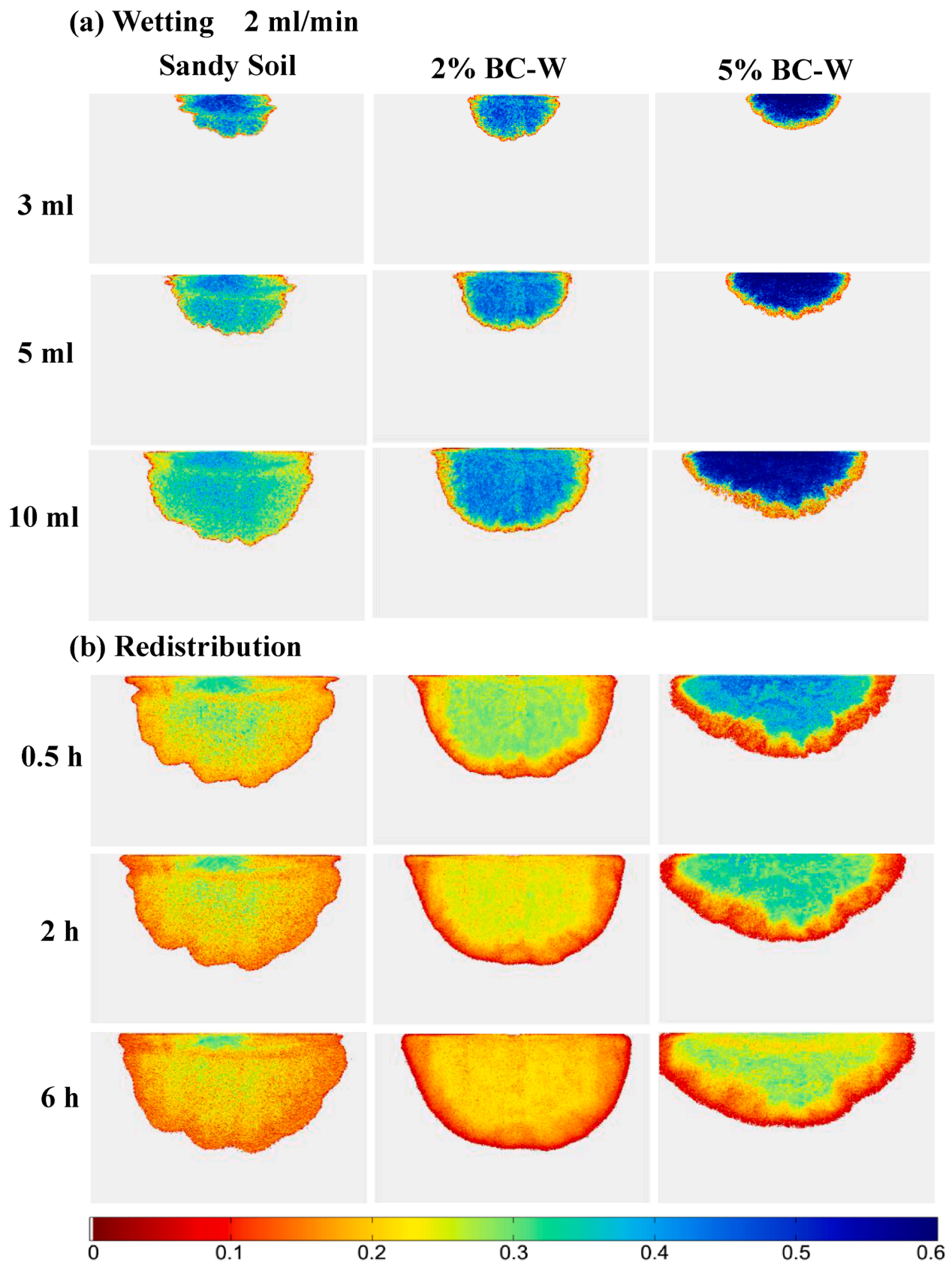


Fig. 7. Soil water content distribution during (a) wetting, and (b) redistribution in sandy soil and soil mixed with 2% and 5% wheat straw derived biochar (BC-W). Water was added to the soil surface as a point source at a flow rate of 2 ml/min. Moisture content varies from red (dry) to blue (wet) according to the color bar.

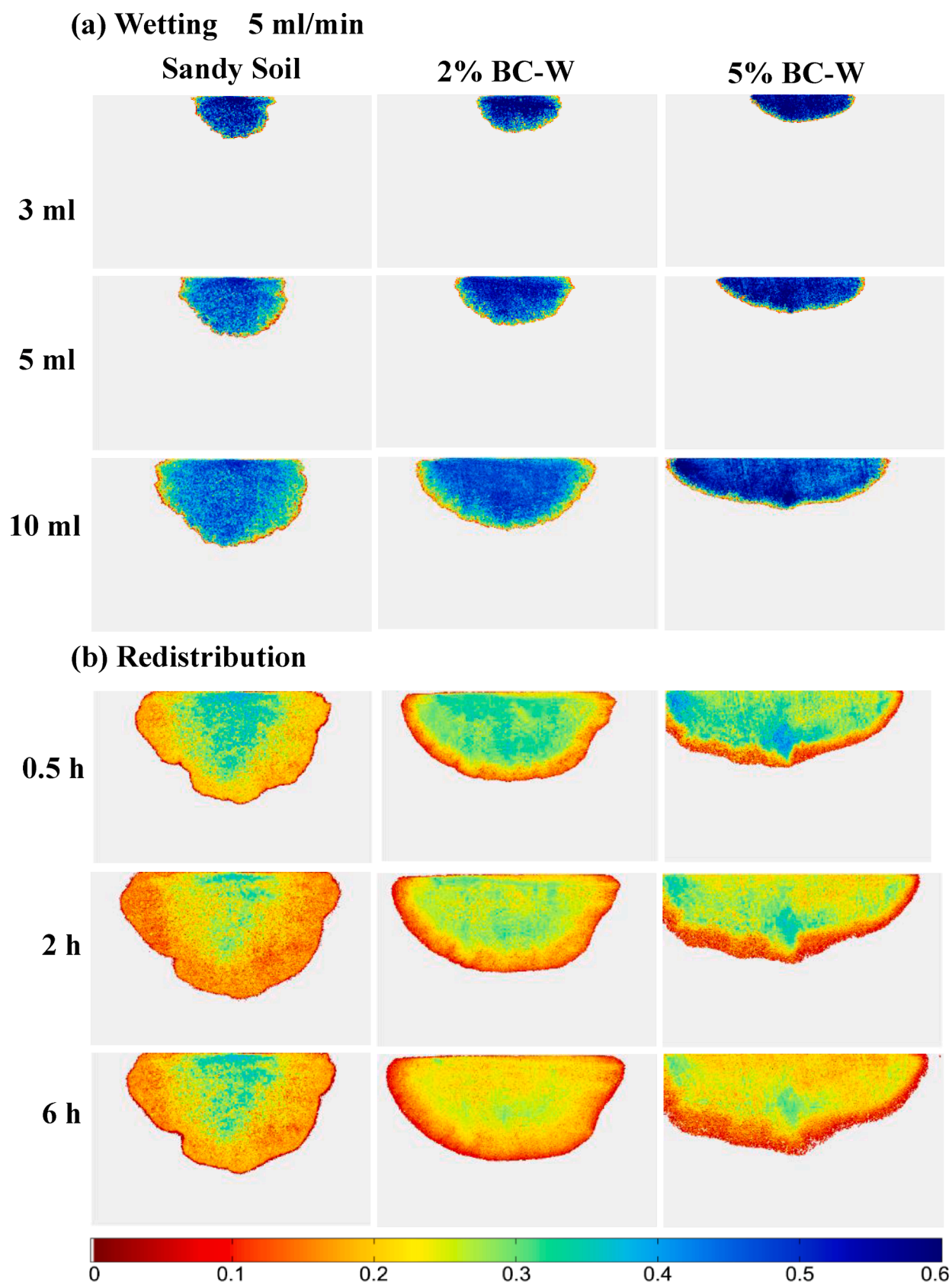


Fig. 8. Soil water content distribution during (a) wetting, and (b) redistribution in sandy soil and soil mixed with 2% and 5% wheat straw derived biochar (BC-W). Water was added to the soil surface as a point source at a flow rate of 5 ml/min. Moisture content varies from red (dry) to blue (wet) according to the color bar.

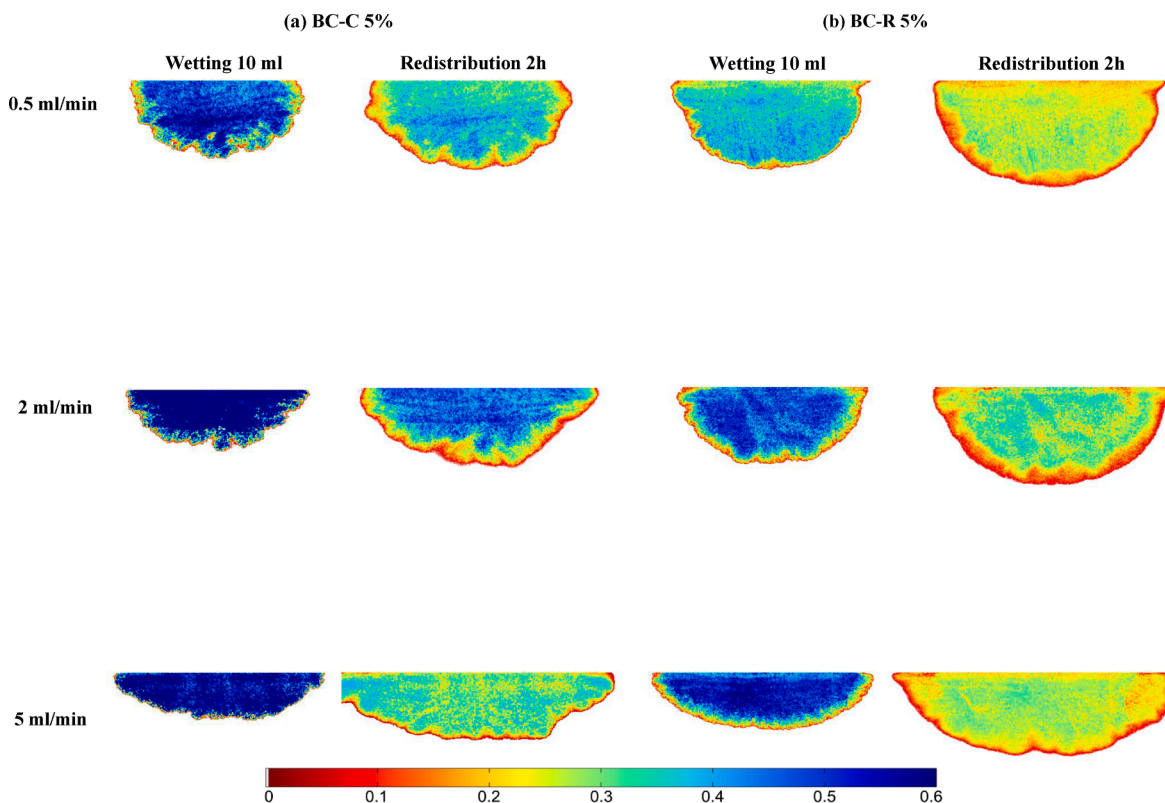


Fig. 9. Soil water content distribution during the end of wetting stage (10 ml), and redistribution stage (2 h) in sandy soil mixed with 5% w/w (a) corn straw derived biochar (BC-C), and (b) rice straw derived biochar (BC-R). Water was added to the soil surface as a point source at a flow rate of 0.5 ml/min, 2 ml/min and 5 ml/min, respectively. Moisture content varies from red (dry) to blue (wet) according to the color bar.

3.4. SWC distribution transects

The SWC distribution patterns within the plumes under different biochar treatments were further investigated by extracting transects along the vertical direction. SWC was taken as the average of the 5 ambient pixels along the transects. Fig. 10 shows the SWC distribution along the vertical transects of the plume at the end of the wetting stage (10 ml water infiltration) and after 2 h redistribution under a 0.5 ml min⁻¹ flow rate in the sandy soil and in the 2% and 5% biochar (BC-W) treatments. In the sandy soil and 2% biochar treatment plumes (Fig. 10a–d), three vertical transects (VT-1, VT-2 and VT-3) were extracted, located in the left, middle (under the dripper) and right parts of the plume, respectively. Given the observed high SWC variability within the plume for the 5% biochar treatment, four vertical transects were taken, with VT-1 and VT-2 for the left part (Fig. 10e,f) and VT-3 and VT-4 for the right part (Fig. 10g,h) of the plume.

At the end of wetting, the SWC distribution along the three transects in the sandy soil plume were similar and showed a general decreasing trend from top to bottom of the plume (Fig. 10a). At the edge of the plume, the SWC decreased sharply from 0.3 cm³ cm⁻³ to 0. After 2 h redistribution, the SWC decreased within the plume while water redistributed laterally and vertically. In addition, the SWC distribution showed a general increasing trend at the bottom of the transects, starting from the location of the wetting front at the end of wetting (~400 pixel depth) (Fig. 10b). In contrast, a clear increasing trend of SWC distribution was obtained from the top to bottom of the plume for the 2% biochar treatment at the end of wetting (Fig. 10c). The highest SWC was located at the bottom of the transects and the SWC decreased gradually from 0.4 cm³ cm⁻³ to 0 at the edge of the plume. After 2 h redistribution, the SWC still showed a gradually increasing trend from plume top to bottom for the 2% biochar treatment (Fig. 10d).

In the 5% biochar treatment plume, the characteristics of the vertical

SWC distribution differed among transect locations. At the end of wetting, a clear increasing trend of SWC was found from the top to bottom of the plume for VT-1, VT-2 (Fig. 10e) and VT-3 (Fig. 10g). Although the SWC along VT-1 and VT-2 had a similar increasing trend and their locations were close to each other, the SWC values were quite different (Fig. 10e). In addition, within the same 5% biochar plume, two contrasting vertical SWC distribution patterns were found (Fig. 10g), with a clear increasing trend along VT-3 and a decreasing trend along VT-4. The results indicated strong spatial heterogeneity of SWC distribution within the plume in the wettable soil mixed with 5% biochar. After 2 h of redistribution, the disparity in SWC along the different transections diminished. However, the increasing trend with vertical depth was still apparent, as shown in VT-2 (Fig. 10f) and VT-4 (Fig. 10h). Similar results were obtained in a replicate experiment using BC-C and BC-R.

4. Discussion

WDPT and CA were used to characterize SWR persistence and intensity, respectively, of three biochars and their mixtures with sandy soil. While the WDPT test indicated no water repellency, the CA measurements demonstrated strong water repellency for the three tested commercial biochars (Figs. 4 and 5). The contradictory results between the two measurement methods can be attributed to the biochar structure that differs substantially from the rigid structure of mineral soils, for which the WDPT test is widely used. While few studies have used the WDPT test to determine biochar water repellency (Baronti et al., 2014), many have used the CA (sessile drop method) for feedstock-derived biochars (i.e., corn stover, pine poplar sludge, hay grass) produced at various pyrolysis temperatures (350–600 °C) (Herath et al., 2013; Jeffery et al., 2015; Ojeda et al., 2015). The WDPT test was found to be affected by the biochar particle size, i.e., it cannot be appropriately performed for biochar ground to pass a 2-mm sieve. Using the WDPT

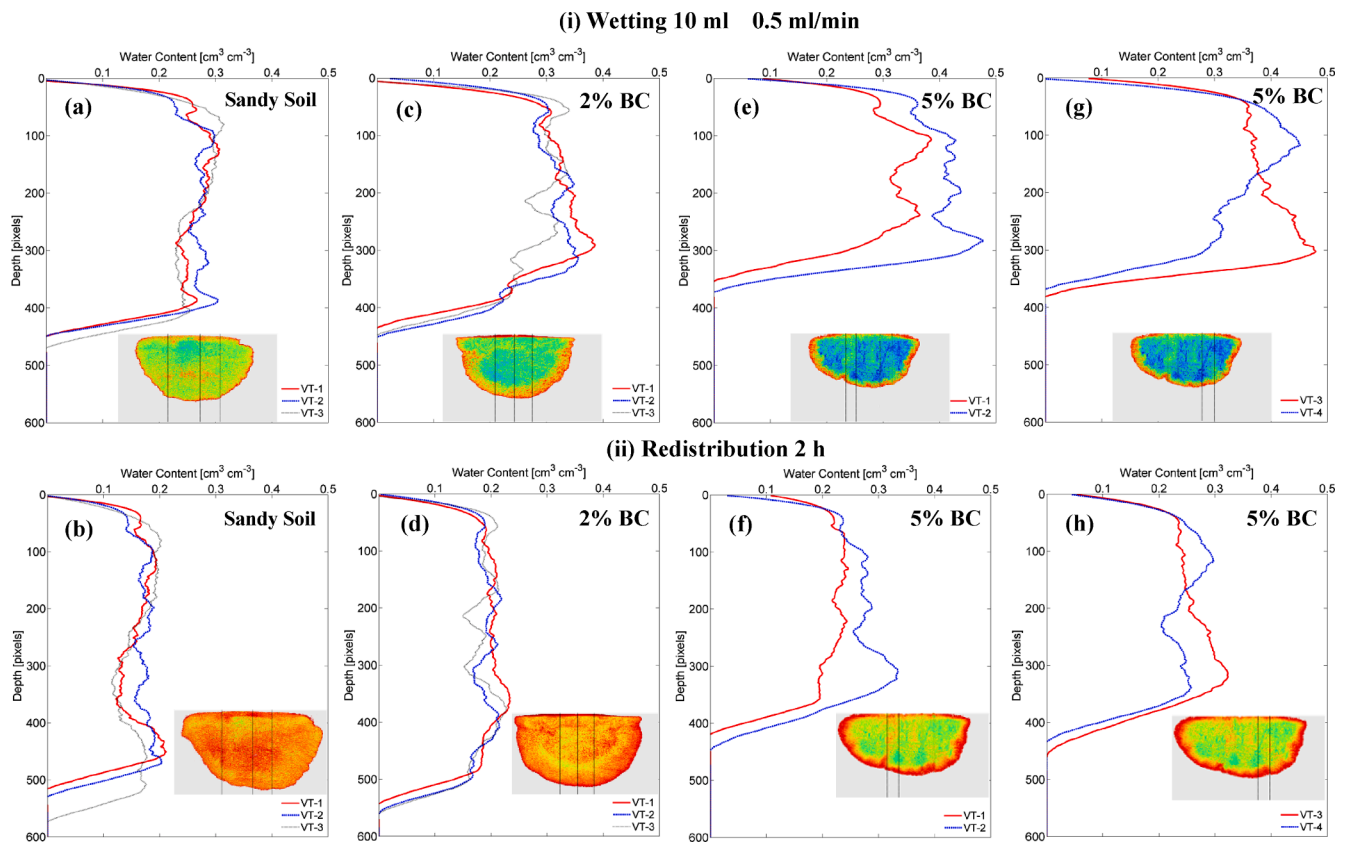


Fig. 10. Soil water content distribution along the vertical transects (VT) of the plumes at the end of (i) wetting stage (10 ml water volume, flow rate of 0.5 ml/min), and (ii) after 2 h redistribution in sandy soil and soil mixed with 2% and 5% wheat straw derived biochar (BC-W).

method, Zornoza et al. (2016) found strong hydrophobicity ($WDPT > 3600$ s) for 4-mm particle size biochar produced from cotton-crop residue at 300 °C. Its common use as a criterion for water repellency on the one hand, and the lack of response of these media to the WDPT test on the other, might explain the common assumption that biochar-soil mixtures are wettable and the lack of studies on the water repellency of biochar-soil mixtures.

The CA, which includes the initial CA ($t = 0$) and its rate of decrease during contact between the soil surface and the water drop, gives physical insight into the wetting dynamics of water-repellent soils. The CA method has been used to quantify wettability effect on soil hydraulic parameters (such as the wetting limb of the pressure-saturation relationship). Therefore, beyond being a screening parameter for water repellency, it has been used in modeling water flow and retention in these media (Brindt and Wallach, 2017, 2020; Wang and Wallach, 2021).

The biochar's hydrophobicity stems from the labile compounds with aliphatic functional groups on the biochar surface (Kinney et al., 2012; Zornoza et al., 2016), as shown by FTIR spectroscopy analysis for the 2800–3000 cm^{-1} C–H stretch region (Fig. 3). Depending on the pyrolysis temperature, aliphatic functional groups can become a critical factor for biochar hydrophobicity. High pyrolysis temperatures (500–700 °C) will promote aromaticity and loss of labile aliphatic C–H functional groups, leading to a decrease in, or even disappearance of water repellency (Gray et al., 2014; Jeffery et al., 2015; Zornoza et al., 2016). In contrast, the presence of aliphatic functional groups at low pyrolysis temperatures (300–400 °C) render the biochar water-repellent. Moreover, biochar produced at low temperatures will have a lower surface area and fewer pores, further decreasing its affinity to water (Das and Sarmah, 2015). The materials studied here were derived from commercial production, where a low pyrolysis temperature (~ 350 – 450 °C) is commonly used to control energy costs, particularly for agriculture. Consequently, a large

proportion of the commercial biochar for land application in agriculture will probably be hydrophobic, and the effects on soil hydraulic properties and water flow should be studied.

Looking at the results obtained for the biochar–soil mixtures using a similar experimental setup that has been used in studies for regular water-repellent soils enables comparing the wetting patterns and dynamics between the two media and relating them to their hydrophobic characteristics. Although the plumes in Figs. 6–8 have a shape similar to those in water-repellent soils (Wallach and Jortzick, 2008; Xiong et al., 2012; Bughici and Wallach, 2016), their SWC distribution is substantially different. While the internal SWC distribution in the water-repellent soil plumes is uniform, it seems to be non-uniform in the biochar–soil plumes. This nonuniformity increases and turns into what seems like internal fingers, in particular for the higher percentage biochar–soil mixtures (Figs. 6 and 7). Moreover, the SWC in these fingers remains high during the redistribution stage and decreases in the soil bulk between them. An additional apparent difference is the roughness of the interface between the wet core and the surrounding drier zone at the bottom of the plume in the biochar–soil mixtures, as opposed to a relatively smooth interface in the water-repellent soils (Wallach and Jortzick, 2008; Xiong et al., 2012; Bughici and Wallach, 2016; Ogunmokin et al., 2020). The rough interface resembles local perturbations that might develop into internal fingers if the experiment included a longer chamber and longer wetting time.

The variation in SWC along the vertical cross-sections at different locations within the plumes for the wetting and following redistribution stages (Fig. 10) provides an additional perspective regarding the biochar effect on SWC distribution in the biochar–soil mixtures. In contrast to the uniform SWC profile in the sandy soil (Fig. 10a,b), the SWC variation in the 2% and 5% biochar–soil mixtures is nonmonotonic (Fig. 10c–h), denoting an increase in SWC with depth, followed by a sharp decrease at the wetting front. This nonmonotonic variation in SWC is associated

with unstable flow and has been dubbed “saturation overshoot” (DiCarlo, 2013). Various parameters and processes at both the micro- and macroscopic scales influence the onset of unstable flow, dictating gravity-induced finger development and influencing their shape and dimensions. Wallach and co-authors (Wallach, 2010; Brindt and Wallach, 2017, 2020; Wang and Wallach, 2021) postulated, and then showed experimentally that unstable flow with a nonmonotonic SWC profile develops during infiltration into media having a non-zero CA between water and its particles. The non-zero CA of the biochar–soil mixtures (Figs. 4 and 5) strengthen that postulation (Figs. 8–10).

Soils rendered water-repellent by wildfire, plants foliage, or effluent irrigation are associated with the coating of mineral soil particles by amphiphilic molecules that gradually turn their hydrophilic part away from the particles while in contact with water (DeBano, 2000; Doerr et al., 2000). In contrast, the current study reveals that the biochar–soil mixtures’ hydrophobicity is caused mainly by the mixing of the hydrophobic biochar particles with soil particles (Figs. 6–8). The hydrophobicity of the well-mixed biochar and sand results from the mixture of coarse biochar particles with the sand and the inclusion of fine biochar particles (Fig. 1) within the pores of the sand particles. It can, therefore, be postulated that the hydrophobicity of the biochar–soil mixtures is physically, rather than chemically induced, the latter being common in water-repellent soils. The moderate decrease in the biochars’ and biochar–soil mixtures’ CA during contact with water reaffirms this postulation. The formation of inner fingers within the major plumes (Figs. 6–9) is, therefore, attributed to the physically induced hydrophobicity of the biochar–soil mixtures. Since there was a negligible change in the particle size distribution for the 2% to 5% mixture (Fig. 2), and given that the bulk density and total porosity were determined for all chamber runs, it may be concluded that the coarse biochar particles control the hydrophobicity of biochar–soil mixtures and the plume dimensions. The latter depends on the biochar percentage in the mixture.

The flow chamber study herein and in previous studies for water repellent soils include water application at the surface of a packed dry soil while following on the spatial and temporal soil water distribution. As known, the SWR is intimately related to SWC. Generally, the SWR decreases with increasing SWC and then suddenly vanishes when the SWC exceeds a critical value (Hermansen et al., 2019; Weber et al., 2021). Thus, a question could be raised regarding the relevancy of these experiments’ outcome for field soils that are not always initially dry. The observed gravity-induced fingering in in-situ studies indicates that such flow regime exists in wet soils as well. The outcome of a recent study by Ogunmukun and Wallach (2021) may provide a clue to this phenomenon. Comparing the repeated time-evolution of the soil electrical resistivity (ER) distribution during irrigation and subsequent drainage in a commercial orchard, for which the soil was rendered water repellent by using treated wastewater for irrigation, revealed that the preferential flow locations pathways in water repellent soil profiles are mainly superimposed. This overlap may indicate the preferential flow pathways’ persistence, implying that a water repellent soil “remembers” the location of preferential flow pathways. The overlapping of the preferential flow pathways after one year observed by the electrical resistance tomography (ERT) scans, independent of the weather conditions on the sampling date, irrigation amounts, and despite the changes in root distribution in the measured layer, leads to a conclusion that a ‘soil memory’ phenomenon is developed in water repellent soils for the preferential flow pathways. Similar future studies in biochar-treated soils rendered water repellent may indicate if a ‘soil memory’ develops in these soils as well.

5. Conclusion

Straw-derived commercial biochars produced under low temperature (350–450 °C) were found to be hydrophobic, having high (> 90°) initial CA. Their addition at ~ 2%–10% by weight to a wettable soil produced a subcritically water-repellent biochar–soil mixture with

significantly increased initial CA (~50°–70°). The high initial CA was persistent, remaining high during the contact time with the water drop. SWR increased with biochar-application rate, whereas the biochar particle size (> 1 mm and < 1 mm) had no significant effect on SWR, indicating that the water repellency of the biochar–soil mixtures is primarily physically, rather than chemically induced. Despite the large number of studies on biochar mixtures with soils, only a few have reported on the water repellency of those mixtures and its effect on water flow. This oversight may be attributed to low WDPT values, which classified these mixtures as wettable. In contrast, the sessile-drop method for CA determination indicated that the biochars and their mixtures with soils are hydrophobic with subcritical repellency, which seems to be a more appropriate method for these cases.

Finger-like plumes developed in the biochar–soil mixtures when water was applied to the soil surface as a point source, simulating a dripper along a drip line. Such fingers indicate that flow in these soils has become unstable. However, as opposed to the uniform SWC in regular water-repellent soils, the inner fingers that formed in the primary plumes in the biochar–soil mixtures could be attributed to the physically induced SWR (blending of the biochar and soil particles) rather than chemically induced SWR (coating of the soil particles with amphiphilic molecules). These inner fingers might develop into autonomous fingers under field conditions, affecting the uniformity of SWC distribution in the biochar–soil mixtures. Such fingers are, in fact, preferential flow paths that induce uneven soil wetting and distribution of fertilizers and salinity in the soil profile (Rahav et al., 2017; Ogunmukun and Wallach, 2021), in turn affecting water and fertilizer availability to the plant roots for plant growth and yield. Further studies should be conducted under field conditions while using ERT scans (Rahav et al., 2017; Ogunmukun and Wallach, 2021) or tracers to investigate whether preferential flow pathways are formed by biochar incorporation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This research was partially supported by The National Natural Science Foundation of China (No. 41771258).

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